Robust Minimax Portfolio Optimization Under Financial Crashes

Matra Acharya

March 9, 2026

Abstract

This project studies the weakness of classical portfolio optimization during financial crisis periods and proposes a more defensive allocation framework using robust minimax optimization. First, the theoretical idea of minimax optimization was learned and understood from the foundational project material. Then historical stock price data for four major Indian stocks—Reliance, ICICI Bank, HDFC Bank, and TCS—were collected. After computing log returns, statistical normality tests were applied to compare market behavior in normal and crash periods. The Jarque–Bera test was used for univariate normality and Mardia's test was used for multivariate normality. The results showed that the return distribution does not satisfy normality assumptions during crisis periods and, more broadly, multivariate normality is not supported for the joint return series. Since the classical Markowitz framework strongly depends on stable means, covariance, and approximate normality, it becomes unreliable during crashes. To address this, a robust minimax portfolio model was developed, where uncertainty in expected returns is explicitly penalized. The project concludes that robust minimax optimization provides a more stable and defensive portfolio allocation under financial stress than classical methods.

1 Introduction

Portfolio optimization is the problem of distributing capital among several assets in such a way that return is improved while risk is controlled. In finance, the classical framework for this problem is the Markowitz mean–variance model. It is elegant, famous, and mathematically clean—which is exactly why reality enjoys kicking it in the shins during crises.

The standard Markowitz framework assumes:

- asset returns are reasonably stable,
- expected returns can be estimated from historical data,

- covariance between assets is meaningful and useful,
- and, at least approximately, return behavior is well-behaved enough for mean–variance analysis.

However, during financial crashes these assumptions start wobbling badly. Extreme losses become more frequent, correlations across assets often increase, and the estimate of expected return becomes noisy and unreliable. A portfolio optimized using fragile assumptions may therefore become highly vulnerable exactly when protection is needed the most.

This project was developed in the following sequence:

1. study the minimax concept and its financial interpretation,
2. collect stock data for Reliance, ICICI, HDFC, and TCS,
3. test whether the return data follow normality assumptions,
4. observe failure of normality during crisis periods,
5. and construct a robust portfolio optimization framework for defensive allocation.

Thus, the central idea of this project is not to predict the exact arrival of a crash, but to design a portfolio that is better prepared when market conditions become adversarial.

2 Objectives

The main objectives of this project are:

- to understand the minimax philosophy in portfolio optimization,
- to test whether stock return data satisfy normality assumptions,
- to compare market behavior in normal and crash periods,
- to explain why the Markowitz method becomes weak during financial crisis,
- and to develop a robust minimax-based portfolio allocation model.

3 Data Description

The study uses four major Indian stocks:

- Reliance Industries,
- ICICI Bank,

- HDFC Bank,
- Tata Consultancy Services (TCS).

The stock prices were downloaded as historical market data and merged by date. The adjusted closing price was used whenever available; otherwise, the closing price was used. After aligning the series, daily log returns were computed.

If $P_{i,t}$ denotes the price of stock i on day t, then the log return is defined as:

$$r_{i,t} = \log\left(\frac{P_{i,t}}{P_{i,t-1}}\right).$$

These returns form the basis for both normality testing and portfolio optimization.

4 Methodology Overview

The methodology of the project has four main steps:

4.1 Step 1: Learning the Minimax Idea

Before constructing the model, the minimax concept was studied theoretically. The minimax philosophy treats the market as if the investor is facing an adverse or worst-case environment. Rather than trusting a single estimated model too strongly, the aim is to choose portfolio weights that remain safer under unfavorable parameter values.

4.2 Step 2: Preparing Stock Return Data

Historical price series were loaded for the four selected stocks. After cleaning the date field and choosing the correct price columns, all data sets were merged using common dates. Then daily log returns were calculated.

4.3 Step 3: Normality Testing

The return data were divided into:

- a normal market period,
- and a crash period.

Then:

- the **Jarque–Bera test** was applied stock-wise,
- and the **Mardia test** was applied jointly to all stock returns.

4.4 Step 4: Portfolio Optimization

After observing the instability of the data distribution, a robust optimization method was used. This model penalizes concentration and uncertainty in expected returns, making the final portfolio more defensive.

5 Normality Testing

5.1 Why Normality Matters

Many classical portfolio models behave better when returns are approximately normal. Under normality, mean and covariance summarize a large part of the distributional behavior. But financial returns during stress periods are rarely so polite. They tend to show:

- skewness,
- heavy tails,
- outliers,
- and changing dependence across assets.

Because of this, it is necessary to test the data before trusting the classical framework.

5.2 Jarque–Bera Test

The Jarque–Bera test checks whether a return series departs from normality by using skewness and kurtosis.

Let:

$$S = \text{sample skewness}, \qquad K = \text{sample kurtosis}, \qquad n = \text{sample size}.$$

Then the Jarque–Bera statistic is:

$$JB = \frac{n}{6}\left(S^2 + \frac{(K-3)^2}{4}\right).$$

Decision rule:

- $p > 0.05$: normality is not rejected,
- $p \leq 0.05$: normality is rejected.

5.3 Mardia Test

The Mardia test checks multivariate normality using two components:

- multivariate skewness,
- multivariate kurtosis.

Suppose $X_1, X_2, \ldots, X_n$ are p-dimensional return vectors, with sample mean $\bar{X}$ and covariance matrix S.

The Mardia multivariate skewness measure is:

$$b_{1,p} = \frac{1}{n^2} \sum_{i=1}^{n} \sum_{j=1}^{n} \left[(X_i - \bar{X})^\top S^{-1} (X_j - \bar{X}) \right]^3 .$$

The Mardia multivariate kurtosis measure is:

$$b_{2,p} = \frac{1}{n} \sum_{i=1}^{n} \left[(X_i - \bar{X})^\top S^{-1} (X_i - \bar{X}) \right]^2 .$$

If either the skewness or kurtosis part shows a small p-value, then multivariate normality is not supported.

6 Results of Normality Testing

The Python results showed the following Jarque–Bera outcomes.

6.1 Jarque–Bera Results

Table 1: Jarque–Bera Test Results

Period	Stock	JB Statistic	p-value	Normal at 5%?
Normal Period	HDFC	447680.9408	0.000000	No
Normal Period	ICICI	56.3249	5.877696×10^{-13}	No
Normal Period	TCS	0.7309	0.6938904	Yes
Normal Period	Reliance	160.1060	1.711716×10^{-35}	No
Crash Period	HDFC	25.8174	2.476424×10^{-6}	No
Crash Period	ICICI	25.0462	3.641465×10^{-6}	No
Crash Period	TCS	11.4057	3.336418×10^{-3}	No
Crash Period	Reliance	34.0268	4.084890×10^{-8}	No

Interpretation: In the normal period, only TCS passed the Jarque–Bera test at the 5% level, while the other three stocks failed. In the crash period, all four stocks failed the test. This indicates that the return behavior becomes clearly non-normal during the crisis period.

6.2 Mardia Test Results

Table 2: Mardia Test Results

Period	Skewness p-value	Kurtosis p-value	Multivariate Normal?
Normal Period	0.000000	0.000000	No
Crash Period	0.022987	4.440892×10^{-16}	No

Interpretation: Both the normal period and the crash period fail the multivariate normality test. The crash-period result is especially important because it confirms that the joint return distribution becomes unsuitable for the usual mean–variance assumptions. In simple words: the market stops behaving like a tidy bell curve and starts behaving like a chaos goblin.

7 Classical Markowitz Portfolio Optimization

7.1 Mean–Variance Framework

Let:

$$\boldsymbol{w} = (w_1, w_2, \ldots, w_n)^\top$$

be the vector of portfolio weights, where w_i is the proportion invested in asset i.

Let:

$$\boldsymbol{\mu} = (\mu_1, \mu_2, \ldots, \mu_n)^\top$$

be the vector of expected returns, and let Σ be the covariance matrix of returns.

Then the expected portfolio return is:

$$R_p = \boldsymbol{w}^\top \boldsymbol{\mu}.$$

The portfolio variance is:

$$\sigma_p^2 = \boldsymbol{w}^\top \Sigma \boldsymbol{w},$$

and the portfolio risk is:

$$\sigma_p = \sqrt{\boldsymbol{w}^\top \Sigma \boldsymbol{w}}.$$

The classical model selects portfolio weights by balancing return and risk. But this method depends heavily on estimated means and covariances being reliable.

7.2 Why Markowitz Struggles in Crashes

During crisis periods:

- expected returns become unstable,
- covariance changes sharply,
- correlation between assets may increase,
- and tail losses become more likely.

Because of this, the optimized portfolio can become misleading. A solution that looks mathematically optimal under calm conditions may perform poorly when the market becomes hostile.

8 Robust Minimax Portfolio Optimization

8.1 Motivation

To reduce sensitivity to unstable mean estimates, a robust optimization approach is used. Instead of assuming that the estimated mean return vector $\hat{\boldsymbol{\mu}}$ is exact, we assume that the true mean lies inside an uncertainty set around it.

8.2 Uncertainty Set for Mean Returns

Let $\hat{\boldsymbol{\mu}}$ be the empirical mean return vector. Then assume:

$$\|\boldsymbol{\mu} - \hat{\boldsymbol{\mu}}\|_2 \leq \delta,$$

where $\delta > 0$ measures the amount of uncertainty in the estimated expected returns.

This means the true mean return may differ from the observed one within a bounded region.

8.3 Minimax Formulation

The robust portfolio problem is written as:

$$\min_{\boldsymbol{w}} \max_{\|\boldsymbol{\mu}-\hat{\boldsymbol{\mu}}\|_2 \leq \delta} \left(-\boldsymbol{w}^\top \boldsymbol{\mu} + \lambda \boldsymbol{w}^\top \hat{\Sigma} \boldsymbol{w} \right),$$

where:

- $\boldsymbol{w}$ is the portfolio weight vector,
- $\hat{\boldsymbol{\mu}}$ is the estimated mean vector,
- $\hat{\Sigma}$ is the estimated covariance matrix,

- δ controls uncertainty in expected returns,
- λ is the risk-aversion parameter.

Using the uncertainty ball, this leads to the robust objective:

$$\min_{\boldsymbol{w}} \left[-\boldsymbol{w}^\top \hat{\boldsymbol{\mu}} + \delta \|\boldsymbol{w}\|_2 + \lambda \boldsymbol{w}^\top \hat{\Sigma} \boldsymbol{w} \right].$$

Subject to:

$$\sum_{i=1}^{n} w_i = 1.$$

If a long-only portfolio is required, then:

$$w_i \geq 0 \qquad \text{for all } i.$$

8.4 Interpretation of the Robust Objective

The objective contains three parts:

1. $-\boldsymbol{w}^\top \hat{\boldsymbol{\mu}}$: encourages higher return,
2. $\delta \|\boldsymbol{w}\|_2$: penalizes concentration and estimation uncertainty,
3. $\lambda \boldsymbol{w}^\top \hat{\Sigma} \boldsymbol{w}$: controls variance-based risk.

Thus, the robust minimax method does not try to forecast the exact crash date. Instead, it builds a portfolio that is less fragile when market estimates become unreliable.

9 Implementation in Python

The implementation followed these main computational steps:

1. load historical stock prices,
2. clean and merge the data on common dates,
3. compute daily log returns,
4. estimate sample mean vector $\hat{\boldsymbol{\mu}}$,
5. estimate covariance matrix $\hat{\Sigma}$,
6. choose robustness parameter δ,
7. choose risk-aversion parameter λ,

8. compute robust weights using:

$$\boldsymbol{w} = (\lambda\hat{\Sigma} + \delta I)^{-1}\hat{\boldsymbol{\mu}},$$

9. normalize weights so that:

$$\sum_{i=1}^{n} w_i = 1.$$

In the Python code, the chosen parameter values were:

$$\delta = 0.1, \qquad \lambda = 1.$$

10 Portfolio Results

The computed portfolio weights from the robust optimization code were approximately:

$$\boldsymbol{w} = \begin{bmatrix} 0.6326 \\ 0.3221 \\ 0.1969 \\ -0.1516 \end{bmatrix}.$$

Interpreting these in the order **Reliance, TCS, ICICI, HDFC**, the model suggests:

- strong positive allocation to Reliance,
- moderate positive allocation to TCS,
- smaller positive allocation to ICICI,
- negative allocation to HDFC.

A negative weight means short exposure if short selling is allowed. If the project intends to remain strictly long-only, then an optimization solver with non-negativity constraints should be used.

The corresponding portfolio quantities were:

$$\text{Portfolio Return} = 0.0010067,$$

$$\text{Portfolio Risk} = 0.0160290,$$

$$\text{Sharpe Ratio} = 0.0628071.$$

For a capital amount of:

$$100000,$$

the suggested allocation becomes approximately:

$$\text{Reliance} = 63262.13,$$

$$\text{TCS} = 32208.87,$$

$$\text{ICICI} = 19691.06,$$

$$\text{HDFC} = -15162.06.$$

11 Discussion

This project shows an important financial idea: a mathematically elegant model is not always a practically safe model.

The Jarque–Bera and Mardia test results indicate that stock returns do not reliably satisfy normality assumptions, especially during crisis periods. Once normality and stability assumptions begin to fail, the classical Markowitz framework becomes more fragile because it depends strongly on the quality of estimated means and covariance.

The robust minimax approach improves this by:

- explicitly allowing uncertainty in expected return estimates,
- discouraging overly concentrated portfolios,
- producing more defensive allocations,
- and reducing sensitivity to unstable market conditions.

This does not mean the robust model is magical. It does not predict the exact crash, eliminate all losses, or fully capture extreme tail risk. But it is a more realistic defensive strategy when market estimates become unreliable.

12 Conclusion

The project successfully combines theoretical understanding and practical implementation. First, the minimax framework was studied conceptually. Then stock return data for Reliance, ICICI, HDFC, and TCS were collected and analyzed. Normality testing showed that the financial return data fail to satisfy classical assumptions, especially during crisis periods. This weakens the foundation of traditional Markowitz optimization.

To overcome this, a robust minimax portfolio model was implemented. By introducing uncertainty in expected returns and penalizing concentration, the model provides a more stable and defensive portfolio allocation. Therefore, robust optimization is better suited than the classical method when markets are under stress.

13 Limitations

This project has some limitations:

- only four stocks were considered,
- transaction costs were not included,
- the model uses a simplified robustness structure,
- tail risk measures such as CVaR were not directly optimized,
- and some weights can become negative if short-selling constraints are not enforced.

14 Future Work

This work can be extended in several directions:

- impose strict long-only constraints,
- include more assets and sector diversification,
- use rolling-window estimation,
- compare with CVaR optimization,
- include stress testing,
- and apply machine learning for market regime detection.